



**BENHA NATIONAL UNIVERSITY
MECHATRONICS ENGINEERING**

**6DOF ROBOT ARM
TRAJECTORY PLANNING REPORT**

Experiment Type: Pick and Place
Date: 2026-06-13

INPUT PARAMETERS

Start	[120.0, 40.0, 180.0]
End	[220.0, 120.0, 140.0]
a2	150.0
a3	130.0

TRAJECTORY

Points: 16

```
[[120.0, 40.0, np.float64(180.0)], [120.046484375, 40.0371875, np.float64(180.02521672573013)],  
[120.184375, 40.1475, np.float64(180.10001837196782)], [120.411328125, 40.3290625,  
np.float64(180.22312558571392)], [120.725, 40.58, np.float64(180.3932373246193)]]
```

JOINT ANGLES ($\theta_1 \rightarrow \theta_6$)

```
[[np.float64(0.3217505543966422), np.float64(-0.6421987857314517),  
np.float64(2.1639446836079452), np.float64(0.04905042891840309),  
np.float64(1.5707963267948966), np.float64(-1.892546881191539)],  
[np.float64(0.3219131778929179), np.float64(-0.6419742433880758),  
np.float64(2.1634452728985623), np.float64(0.04932529728441019),  
np.float64(1.5707963267948966), np.float64(-1.8927095046878144)],  
[np.float64(0.32239473847869854), np.float64(-0.6413078986990577),  
np.float64(2.1619632978653174), np.float64(0.050140927628636964),  
np.float64(1.5707963267948966), np.float64(-1.893191065273595)],  
[np.float64(0.323184598153241), np.float64(-0.6402103437060209),  
np.float64(2.159522412130147), np.float64(0.05148425837077065),  
np.float64(1.5707963267948966), np.float64(-1.8939809249481376)],  
[np.float64(0.3242706915798259), np.float64(-0.6386918635894592),  
np.float64(2.1561453913690083), np.float64(0.053342799015347546),  
np.float64(1.5707963267948966), np.float64(-1.8950670183747225)]]
```

JACOBIAN ANALYSIS

Samples: 16

```
[[ 3.03638402e+01 -2.51039521e-14 -8.52331654e+01 3.79473319e+01 -1.26491106e+01  
1.30068164e-15] [ 1.43454613e+02 -2.03672023e-16 -2.84110551e+01 1.26491106e+01  
3.79473319e+01 2.97857438e-15] [-0.00000000e+00 1.26491106e+02 6.37399913e+00  
7.10542736e-15 -3.37040013e-15 -2.64550549e-31] [ 0.00000000e+00 3.16227766e-01  
3.16227766e-01 3.16227766e-01 9.48683298e-01 7.44643595e-17] [ 0.00000000e+00  
-9.48683298e-01 -9.48683298e-01 -9.48683298e-01 3.16227766e-01 -1.21334883e-16] [  
1.00000000e+00 6.12323400e-17 6.12323400e-17 6.12323400e-17 1.51254998e-18  
-1.00000000e+00]]
```

SUMMARY

This report demonstrates trajectory planning for a 6DOF robotic arm. It includes inverse kinematics, forward kinematics, and Jacobian analysis for motion validation and singularity detection.